ISE 555 HW 5

Luke Pepin

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Solution sheet: `HW5.xlsx` (tableaus for Questions 1 and 4). Pivots are boxed, and the row 0 RHS is $-z$.

Question 1

The first constraint is $\geq$, so it gets a surplus e_1 and an artificial a_1. The second is an equality, so it gets an artificial a_2.

Minimize: $z = -4x_1 - 2x_2 + Ma_1 + Ma_2$

Subject to:

- $3x_1 - 2x_2 - e_1 + a_1 = 4$
- $x_1 - 2x_2 + a_2 = 2$
- $x_1, x_2, e_1, a_1, a_2 \geq 0$

a_1 and a_2 start in the basis, so row 0 is priced out by subtracting M(row 1) and M(row 2) from the cost row.

Iteration 0

	x_1	x_2	e_1	a_1	a_2	RHS	ratio
row 0	$-4-4M$	$-2+4M$	M	0	0	$-6M$	
row 1 (a_1)	$\boxed{3}$	-2	-1	1	0	4	4/3
row 2 (a_2)	1	-2	0	0	1	2	2

x_1 enters, a_1 leaves.

Iteration 1

	x_1	x_2	e_1	a_1	a_2	RHS	ratio
row 0	0	$-14/3+4M/3$	$-4/3-M/3$	$4/3+4M/3$	0	$16/3-2M/3$	
row 1 (x_1)	1	$-2/3$	$-1/3$	$1/3$	0	4/3	–
row 2 (a_2)	0	$-4/3$	$\boxed{1/3}$	$-1/3$	1	2/3	2

e_1 enters, a_2 leaves.

Iteration 2

	x_1	x_2	e_1	a_1	a_2	RHS	ratio
row 0	0	-10	0	M	$4+M$	8	
row 1 (x_1)	1	-2	0	0	1	2	–
row 2 (e_1)	0	-4	1	-1	3	2	–

x_2 enters (reduced cost -10), but its column has no positive entries, so there is no ratio test.

Answer: Unbounded, no optimal solution ($z \to -\infty$). Both artificials have left the basis, so the problem is feasible.

Question 2

Standard form

Minimize: $-z = -2x_1 + 3x_2$

Subject to:

- $x_1 + x_2 - e_1 = 3$
- $3x_1 + x_2 + s_2 = 6$
- $x_1, x_2, e_1, s_2 \geq 0$

Canonical primal form

The $\leq$ constraint is multiplied by -1 so that every constraint is $\geq$.

Minimize: $-z = -2x_1 + 3x_2$

Subject to:

- $x_1 + x_2 \geq 3$
- $-3x_1 - x_2 \geq -6$
- $x_1, x_2 \geq 0$

Dual form

The canonical primal is minimize $c^T x$ subject to $Ax \geq b$, $x \geq 0$, with

$$c = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ -3 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ -6 \end{bmatrix}.$$

Its dual is maximize $w = b^T y$ subject to $A^T y \leq c$, $y \geq 0$, with one y_i for each primal constraint and

$$A^T = \begin{bmatrix} 1 & -3 \\ 1 & -1 \end{bmatrix}.$$

Maximize: $w = 3y_1 - 6y_2$

Subject to:

- $y_1 - 3y_2 \leq -2$ (x_1 column)
- $y_1 - y_2 \leq 3$ (x_2 column)
- $y_1, y_2 \geq 0$

Question 3

The primal is minimize $c^T x$ subject to $Ax = b$, $x \geq 0$, with

$$c = \begin{bmatrix} -2 \\ 4 \\ -3 \end{bmatrix}, \quad A = \begin{bmatrix} 9 & -2 & -8 \\ 3 & 3 & 3 \\ 7 & -5 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}.$$

Each equality can be written as a pair of $\geq$ constraints, $Ax \geq b$ and $-Ax \geq -b$, with dual variables $u, v \geq 0$. They only appear as $y = u - v$, so each y_i is free. The dual is maximize $w = b^T y$ subject to $A^T y \leq c$, y free, with

$$A^T = \begin{bmatrix} 9 & 3 & 7 \\ -2 & 3 & -5 \\ -8 & 3 & 2 \end{bmatrix}.$$

Maximize: $w = 5y_1 + 7y_2 + 9y_3$

Subject to:

- $9y_1 + 3y_2 + 7y_3 \leq -2$ (x_1 column)
- $-2y_1 + 3y_2 - 5y_3 \leq 4$ (x_2 column)
- $-8y_1 + 3y_2 + 2y_3 \leq -3$ (x_3 column)
- y_1, y_2, y_3 free (unrestricted in sign)

Question 4

Minimize: $z = -x_1 + 2x_2 - x_3$

Subject to:

- $x_1 + 2x_2 + x_3 + s_1 = 12$
- $2x_1 + x_2 - x_3 + s_2 = 6$
- $-x_1 + 3x_2 + s_3 = 9$
- $x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$

Iteration 0

	x_1	x_2	x_3	s_1	s_2	s_3	RHS	ratio
row 0	−1	2	−1	0	0	0	0	
row 1 (s_1)	1	2	1	1	0	0	12	12
row 2 (s_2)	$\boxed{2}$	1	−1	0	1	0	6	3
row 3 (s_3)	−1	3	0	0	0	1	9	–

x_1 and x_3 tie at -1; x_1 enters, s_2 leaves.

Iteration 1

	x_1	x_2	x_3	s_1	s_2	s_3	RHS	ratio
row 0	0	2.5	−1.5	0	0.5	0	3	
row 1 (s_1)	0	1.5	$\boxed{1.5}$	1	−0.5	0	9	6
row 2 (x_1)	1	0.5	−0.5	0	0.5	0	3	–
row 3 (s_3)	0	3.5	−0.5	0	0.5	1	12	–

x_3 enters, s_1 leaves.

Iteration 2

	x_1	x_2	x_3	s_1	s_2	s_3	RHS
row 0	0	4	0	1	0	0	12
row 1 (x_3)	0	1	1	2/3	$-1/3$	0	6
row 2 (x_1)	1	1	0	1/3	1/3	0	6
row 3 (s_3)	0	4	0	1/3	1/3	1	15

Row 0 is all ≥ 0, so this is optimal.

Answer: $x_1 = 6$, $x_2 = 0$, $x_3 = 6$, $z = -12$

Dual form

Each $\leq$ constraint is multiplied by -1 to get the canonical form, minimize $c^T x$ subject to $Ax \geq b$, $x \geq 0$, with

$$c = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & -2 & -1 \\ -2 & -1 & 1 \\ 1 & -3 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -12 \\ -6 \\ -9 \end{bmatrix}.$$

The dual is maximize $w = b^T y$ subject to $A^T y \leq c$, $y \geq 0$, with

$$A^T = \begin{bmatrix} -1 & -2 & 1 \\ -2 & -1 & -3 \\ -1 & 1 & 0 \end{bmatrix}.$$

Maximize: $w = -12y_1 - 6y_2 - 9y_3$

Subject to:

- $-y_1 - 2y_2 + y_3 \leq -1$ (x_1 column)
- $-2y_1 - y_2 - 3y_3 \leq 2$ (x_2 column)
- $-y_1 + y_2 \leq -1$ (x_3 column)
- $y_1, y_2, y_3 \geq 0$

Complementary slackness

If a primal variable is positive, its dual constraint is tight. If a primal constraint has slack, its dual variable is 0. The primal optimum has $x_1 = 6$, $x_2 = 0$, $x_3 = 6$, $s_1 = 0$, $s_2 = 0$, $s_3 = 15$.

Primal	Dual
$s_3 = 15 > 0$	$y_3 = 0$
$x_1 = 6 > 0$	$-y_1 - 2y_2 + y_3 = -1$
$x_3 = 6 > 0$	$-y_1 + y_2 = -1$
$x_2 = 0$	no information

With $y_3 = 0$, subtracting the third equation from the second gives $-3y_2 = 0$, so $y_2 = 0$ and $y_1 = 1$.

Check: the x_2 constraint gives $-2(1) - 0 - 0 = -2 \leq 2$, and $y \geq 0$, so y is dual feasible. Also $w = -12(1) = -12 = z^*$. This matches row 0 of the final tableau under s_1, s_2, s_3.

Answer: $y_1 = 1$, $y_2 = 0$, $y_3 = 0$, $w = -12$

Question 5

Dual problems ask, "What is each resource worth?" In HW3 we used optimization to decide which of 3 factories to close. If each factory's capacity is a constraint, its dual value (shadow price) is how much profit the company loses per unit of that capacity. A factory with slack has a dual value of 0. It is not a bottleneck,

so it is the best candidate to close. A factory with a high dual value is limiting profit and should stay. Shadow prices only hold for small changes, and closing a factory removes all of its capacity, so we re-solve the model without that factory to confirm. The dual tells us where to look, and re-solving gives the answer.